

2. Image 2.1 shows chickens in two different farming systems.

Image 2.1



Free-range farm

Intensive farm

In 1950 chicken was an expensive food and most adults in the UK, on average, ate only 1100 g each year. By 2000 they ate 25 kg per year as intensive farming had made chicken much cheaper.

In intensive farming, large numbers of chickens are reared indoors. Environmental conditions and food supply are constantly controlled. The chickens grow faster than free-range chickens and use less energy as their movement is restricted.

The farmer can monitor the chickens more easily than on a free-range farm. Much less land is used and labour costs are lower but larger amounts of concentrated waste are produced.

- (a) (i) Use the information given to calculate the increase in mass in the annual consumption of chicken for a **family of four adults** in the UK between 1950 and 2000. [2]

Increase in mass = kg/family/year



During an investigation, the growth of chickens from different farm systems was compared. The results are shown in **Graph 2.2** and **Tables 2.3** and **2.4**.

Graph 2.2

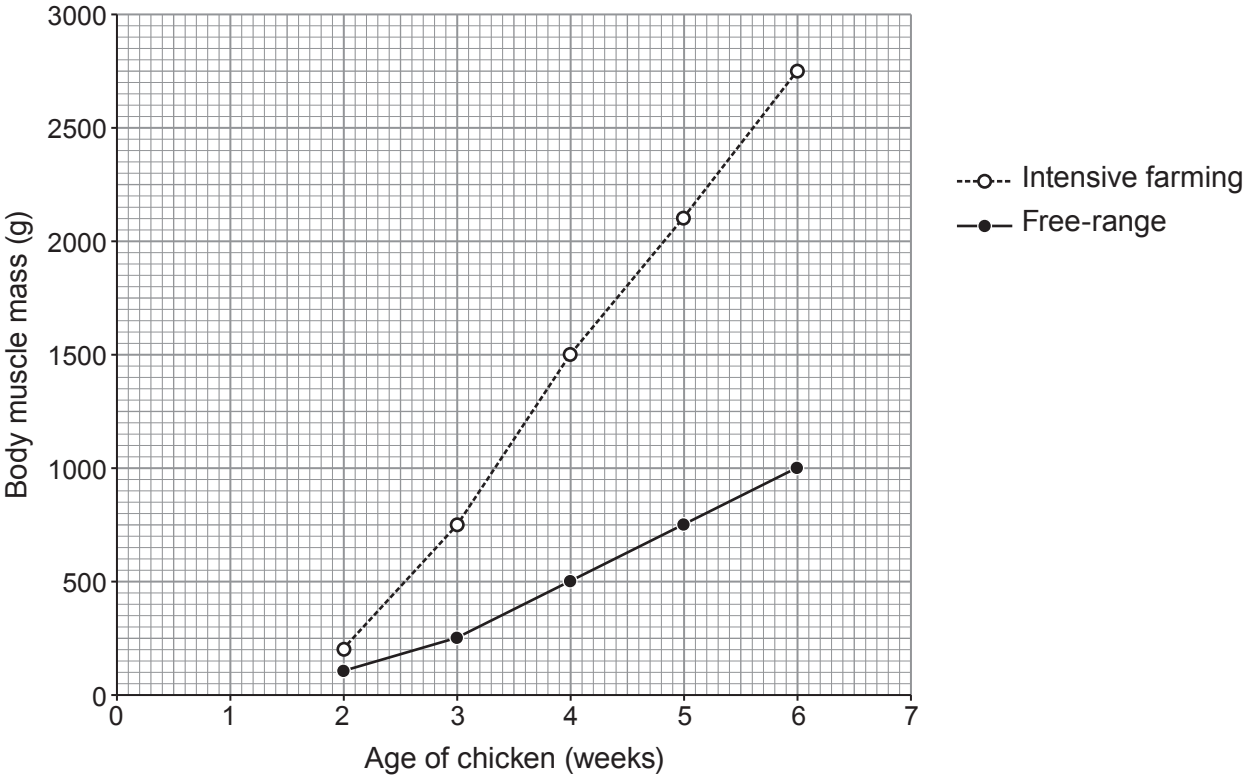


Table 2.3

Organ mass in chickens at six weeks

Organ	Mass (g)	
	Free-range	Intensively farmed
Heart	6.5	4.8
Lungs	4.0	3.5
Liver	20.0	15.0

Table 2.4

Bone quality in chickens at six weeks

	Free-range	Intensively farmed
Bone density (g/cm ³)	1.29	0.79
Presence of broken bones (%)	2.5	37
Length of leg bones (mm)	73	118



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- (ii) Use **Graph 2.2** to calculate the difference in the body muscle mass at 5 weeks between intensively farmed chicken and a free-range chicken. [1]

Difference = g

- (b) (i) Use the information on page 7 to suggest **one** advantage to farmers of farming chickens intensively. [1]

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- (ii) State **one** feature of intensive farming which is an environmental disadvantage. [1]

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- (c) Some groups of people have ethical objections to intensive farming because of animal welfare concerns.

Using the information in **Tables 2.3** and **2.4**, state **three** features of intensive farming which support this point of view. [3]

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